

COURSE: EEE 4484 (DIGITAL ELECTRONICS & PULSE TECHNIQUES LAB.)
EXPERIMENT NO. 02
STUDY OF RTL NOR GATE & TTL NAND GATE

OBJECTIVE:

1. Study of RTL (Resistor Transistor Logic) NOR gate.
2. To analyze the circuits of TTL (Transistor-Transistor Logic) NAND gate with totem-pole output.

INTRODUCTION:

Resistor–transistor logic (RTL) is a class of digital circuits built using resistors as the input network and bipolar junction transistors (BJTs) as switching devices. RTL circuit was succeeded by diode–transistor logic (DTL) and transistor–transistor logic (TTL). Transistor–transistor logic (TTL) is a logic family built from bipolar junction transistors in which transistors perform both the logic function and the amplifying function.

EQUIPMENT:

1. DC Power Supply,
2. Breadboard,
3. Resistors (5 Nos),
4. Diode (1 Nos),
5. Transistors (BJTs) (5 Nos),
6. Multimeter, and
7. Connecting wires.

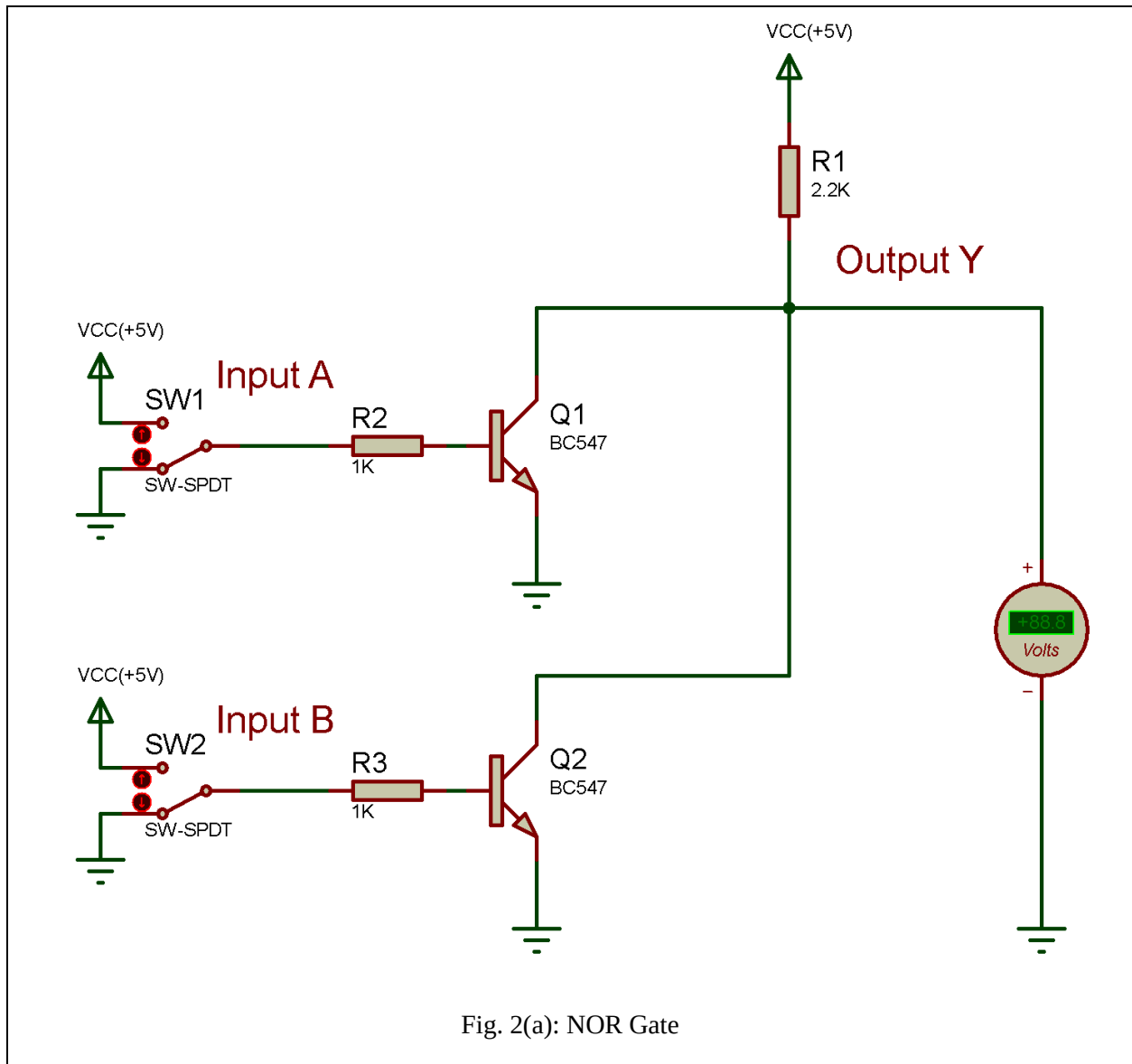
KNOWLEDGE PROFILE AND APPLICATION:

1. Operation of Universal Gates (NAND and NOR),
2. Totem Pole Output Configuration, and
3. Common Emitter Configuration of BJT.

PROCEDURE:

1. Connect the following circuits given in Fig. 2(a) and Fig. 2(b) in breadboard.
2. Measure the output voltage using multimeter and fill up the corresponding truth tables.
3. Use +5V power supply from the board to connect the switches.

CIRCUIT DIAGRAM & TRUTH TABLE:



Data Table:

Input A	Input B	Output Voltage (Vo)	Logic Output Level

**Data Table:**

REPORT QUESTION:

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